

ROUTE 53-DomainRouting

Prepared By: Kritika Agarwal

Project: Route 53-Domain Routing

Objective

- Login AWS
- Project purpose

Architecture Diagram

- Structure
- Explanation

Aws services used

- Amazon Route 53
- Amazon S3
- Amazon EC2
- Elastic IP
- VPC
- Security Group

Testing Outcome

- Upload successfully
- Created successfully

Conclusion

- Project summary
- Learning Outcome

1. OBJECTIVE-

- Purpose: Login to AWS Management Console using IAM User or Root User Credentials
Store files Securely.

Steps:

1. Login to AWS Management Console.
2. Search Amazon S3 Bucket in the search bar.
3. Open the Amazon S3 Bucket Dashboard.

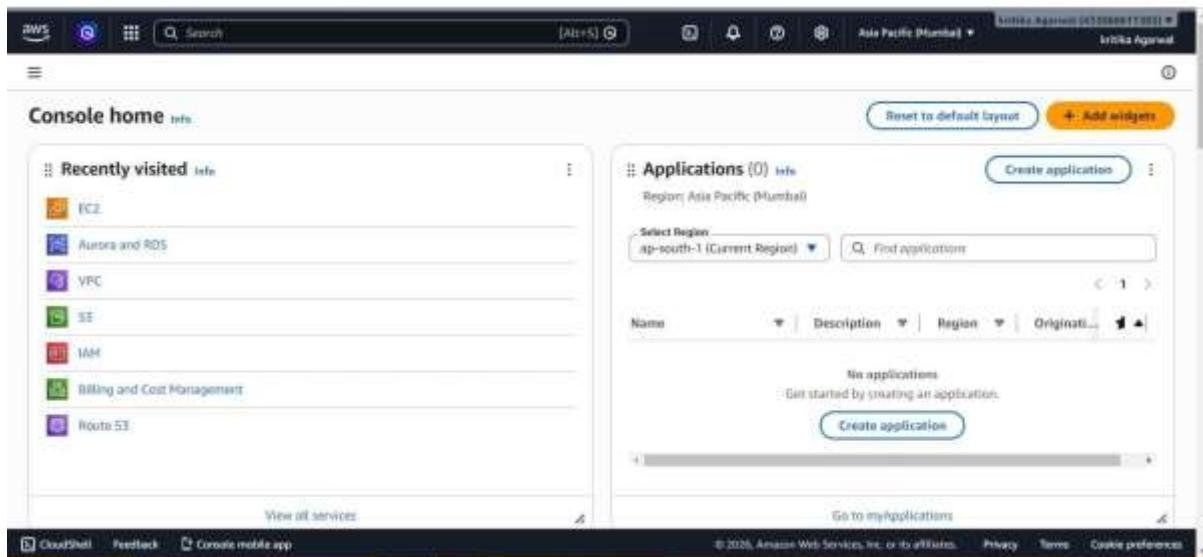


FIG: AWS CONSOLE HOME PAGE

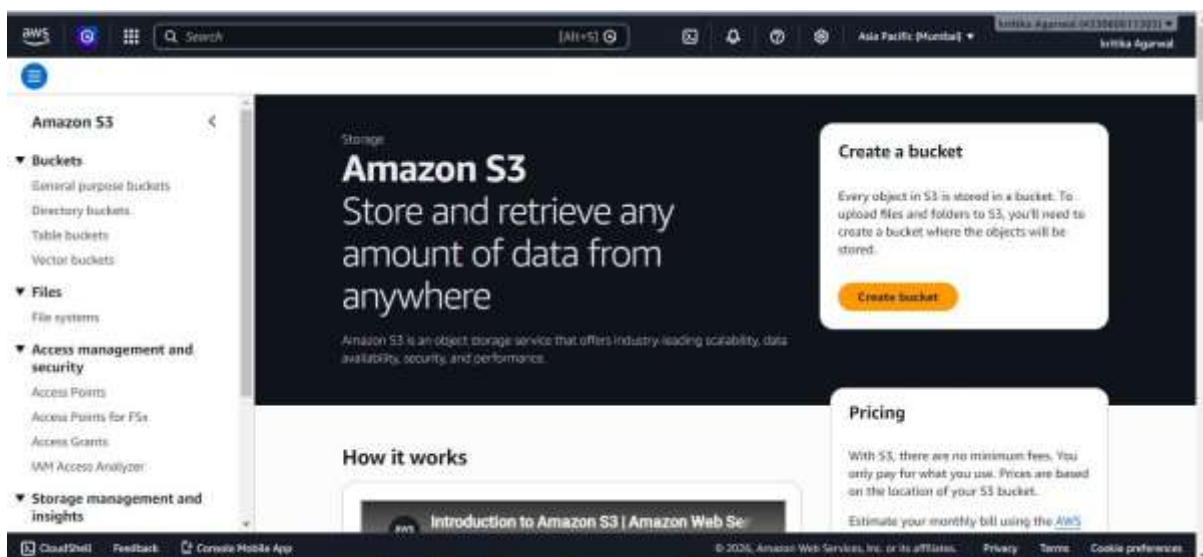


FIG: S3 DASHBOARD

2. ARCHITECTURE DIAGRAM-

- Internet users custom domain ke through website access karte hain.
- Domain requests ko Amazon Route 53 manage karta hai.
- Alias Record traffic ko Amazon S3 Static Website par route karta hai.
- A Record traffic ko EC2 ki Elastic IP par route karta hai.
- S3 bucket static website host karta hai, jabki EC2 Apache/Nginx ke through live website serve karta hai.
- Is architecture se ek hi domain se static aur dynamic website hosting ko efficiently manage kiya ja sakta hai.

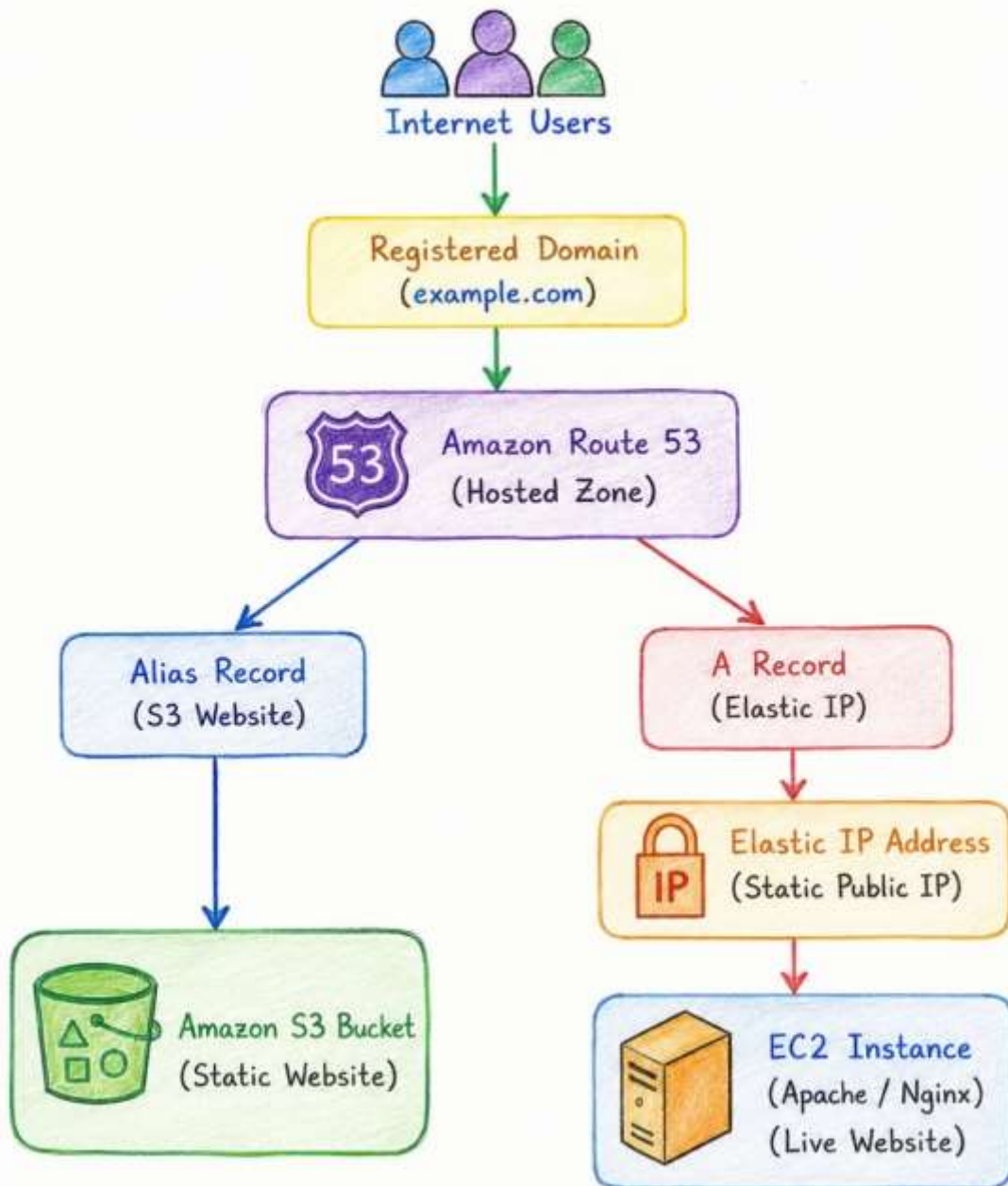


FIG: ROUTE 53

3. AWS SERVICES USED-

PART-1: S3 Static Website with Route 53:

➤ Step 1- Create S3 Bucket:

- Open AWS Console
- Go to S3
- Click Create Bucket
- Bucket Name : kritika-static-website-2026

- Region select karo
- Bucket create karo

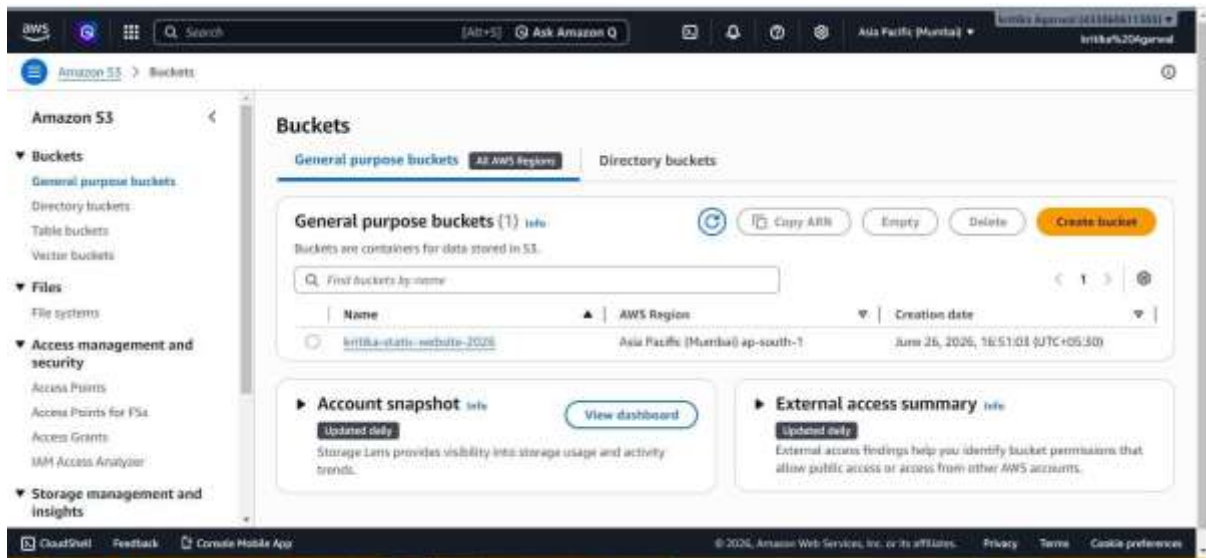


FIG: BUCKET NAME

➤ Step 2-Enable Static Website Hosting:

- Bucket open karo
- Properties
- Static Website Hosting
- Enable
- Index Document
- Error Document



FIG: ENABLE WEBSITE

➤ Step 3-Upload Website Files:

- Upload
- index.html
- error.html

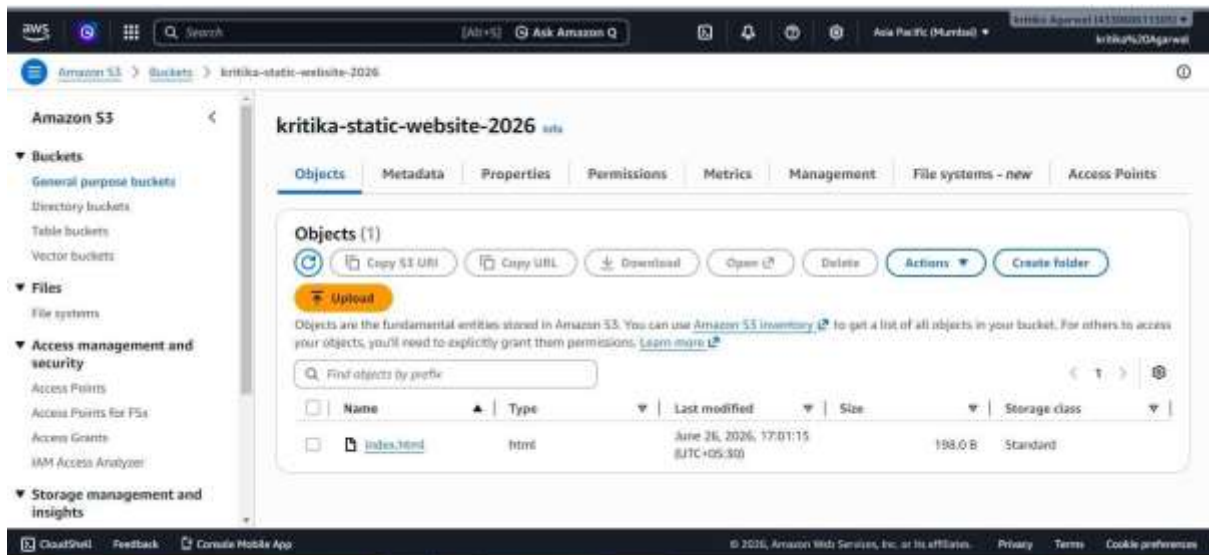


FIG: UPLOAD FILES

➤ Step 4-Disable Block Public Access:

- permission
- Disable: Block All Public Access
- Save Changes

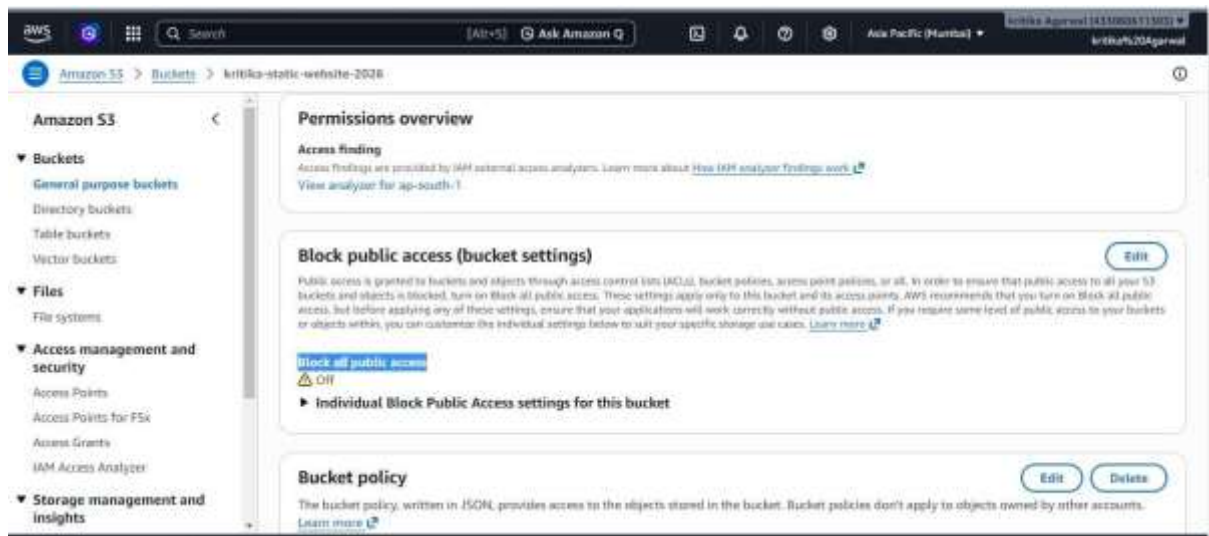


FIG: BLOCK ACCESS

➤ Step 5-Bucket Policy:

- Bucket Policy add it.

➤ Step 6-Copy Website Endpoint:

- SWH Copy.

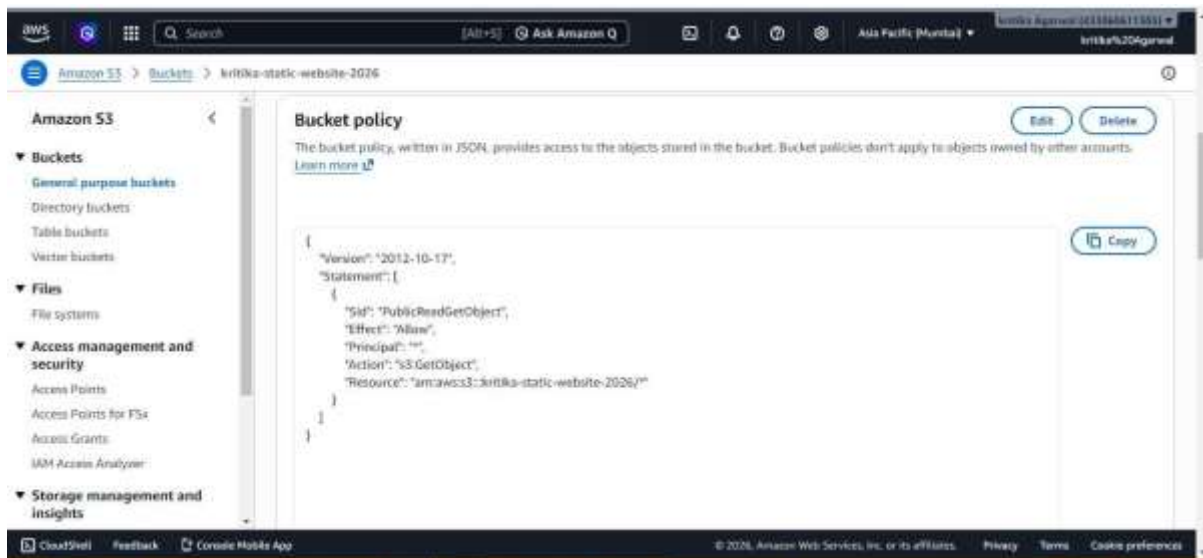


FIG-1: BUCKET POLICY



FIG-2: WEBSITE ENDPOINT

➤ Step 7-Route 53 Hosted Zone:

- AWS Console
- Route 53
- Hosted Zones
- Select Domain

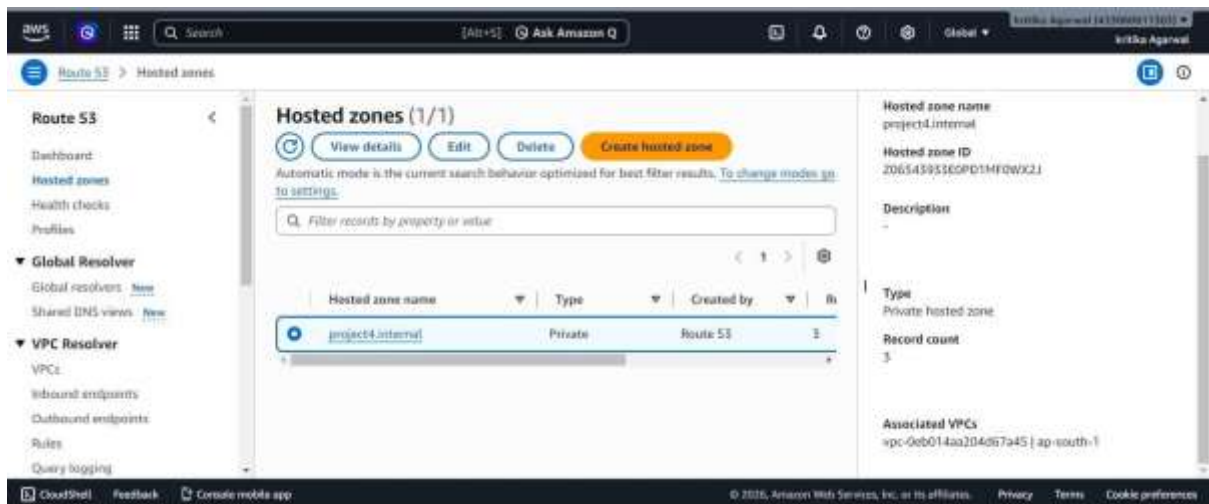


FIG: HOSTED ZONE

➤ Step 8-Create Alias Record:

- Create Record
- Type:A
- Alias:Yes
- Alias Target:S3 Website Endpoint
- Bucket Select: kritika-static-website-2026
- Save

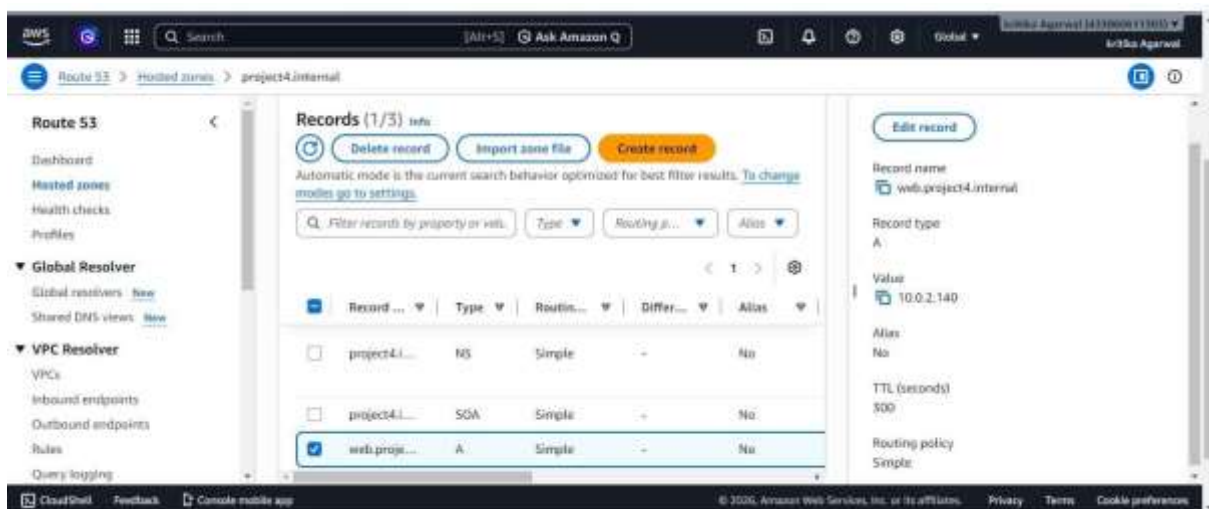


FIG: ALIAS RECORD

Step 9-DNS Propagation:

- Wait
- 5-10 minutes

Step 10-Verification:

- Browser: www.yourdomain.com
- Output: Static Website Successfully Opened.



FIG: WEBSITE OPEN SUCCESSFULLY

4. TESTING OUTCOME:

PART-2: EC2 Live Website With Route 53:

➤ Step 1-Launch EC2:

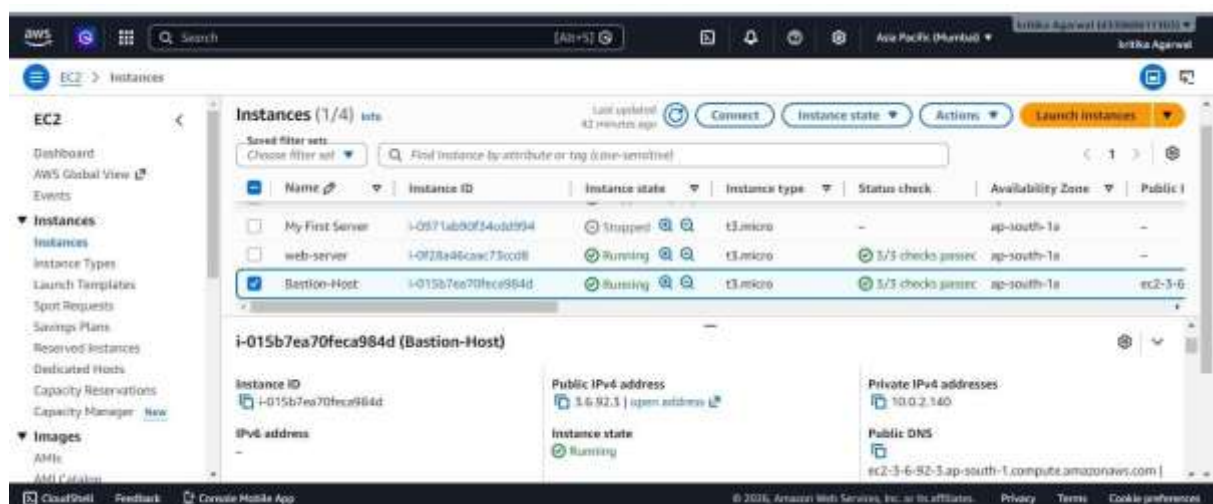
❖ Configuration

- Amazon Linux 2023
- T3.micro
- Key Pair : set-key

❖ Security Group: inbound rules-

- Port 22 (SSH)
- Port 80 (HTTP)
- Port 443 (HTTPS)

FIG: EC2 LAUNCH PAGE



➤ Step 2-Connect EC2:

- `ssh -i key.pem ec2-user@Public-IP`
- SSH Connection.

```
Windows PowerShell
Copyright (C) 2014 Microsoft Corporation. All rights reserved.

PS C:\Users\Mohit> cd downloads
PS C:\Users\Mohit\downloads> ssh -i "set.pem" ec2-user@ec2-3-6-92-3.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-3-6-92-3.ap-south-1.compute.amazonaws.com (3.6.92.3)' can't be established.
ECDSA key fingerprint is SHA256:o0QvEm2s659WGOfgV98Mui+XoLFiXRudw1+EZ9A+Ads.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-6-92-3.ap-south-1.compute.amazonaws.com' (ECDSA) to the list of known hosts.
```



```
#
#####
#####
#####
#
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Fri Jul 10 16:03:30 2026 from 157.48.98.29
[ec2-user@ip-10-0-2-140 ~]$ sudo -i
[root@ip-10-0-2-140 ~]# sudo systemctl status httpd

■ httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/sbin/httpd -DFOREGROUND
Jul 10 16:05:47 ip-10-0-2-140.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jul 10 16:05:47 ip-10-0-2-140.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jul 10 16:05:47 ip-10-0-2-140.ap-south-1.compute.internal httpd[26218]: Server configured, listening on: port 80

sudo yum install httpd -y

Last metadata expiration check: 0:26:06 ago on Fri Jul 10 16:04:39 2026.
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-10-0-2-140 ~]# sudo systemctl enable httpd
[root@ip-10-0-2-140 ~]# sudo systemctl start httpd

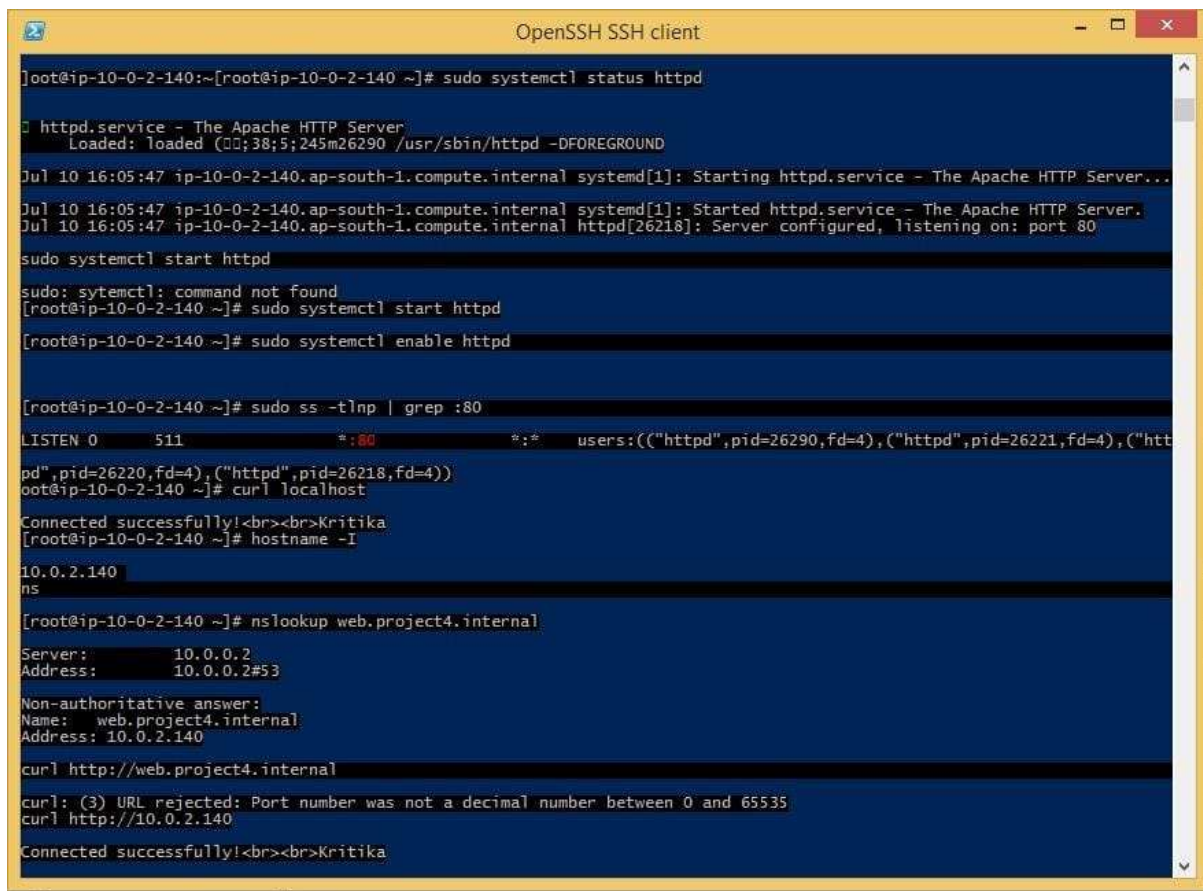
oot@ip-10-0-2-140 ~]# echo "Connected successfully!  
  
Kritika" | sudo tee /var/www/html/index.html

-bash: !: event not found
[root@ip-10-0-2-140 ~]# sudo sh -c 'echo "Connected successfully!  
  
Kritika" > /var/www/html/index.html'
```

FIG: SSH CONNECTION

➤ Step 3- Install Apache:

- Sudo yum update -y
- Sudo yum install httpd -y
- sudo systemctl start httpd
- sudo systemctl enable httpd
- ❖ Check-
- Systemctl status httpd



```
[root@ip-10-0-2-140:~][root@ip-10-0-2-140 ~]# sudo systemctl status httpd
■ httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; vendor preset: enabled)
   Active: active (running) since Jul 10 16:05:47; 1min 1s ago
     Main PID: 26218 (httpd)
       CGroup: /systemd/system/httpd.service
               └─ 26218 /usr/sbin/httpd -DFOREGROUND

Jul 10 16:05:47 ip-10-0-2-140.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jul 10 16:05:47 ip-10-0-2-140.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jul 10 16:05:47 ip-10-0-2-140.ap-south-1.compute.internal httpd[26218]: Server configured, listening on: port 80

[root@ip-10-0-2-140 ~]# sudo systemctl start httpd
sudo: systemctl: command not found
[root@ip-10-0-2-140 ~]# sudo systemctl start httpd
[root@ip-10-0-2-140 ~]# sudo systemctl enable httpd

[root@ip-10-0-2-140 ~]# sudo ss -tlnp | grep :80
LISTEN 0      511          *:*          *:*    users:((("httpd",pid=26290,fd=4),("httpd",pid=26221,fd=4),("httpd",pid=26220,fd=4),("httpd",pid=26218,fd=4)))
[root@ip-10-0-2-140 ~]# curl localhost
Connected successfully!<br><br>Kritika
[root@ip-10-0-2-140 ~]# hostname -I
10.0.2.140
ns
[root@ip-10-0-2-140 ~]# nslookup web.project4.internal
Server:      10.0.0.2
Address:     10.0.0.2#53

Non-authoritative answer:
Name:   web.project4.internal
Address: 10.0.2.140

curl http://web.project4.internal
curl: (3) URL rejected: Port number was not a decimal number between 0 and 65535
curl http://10.0.2.140
Connected successfully!<br><br>Kritika
```

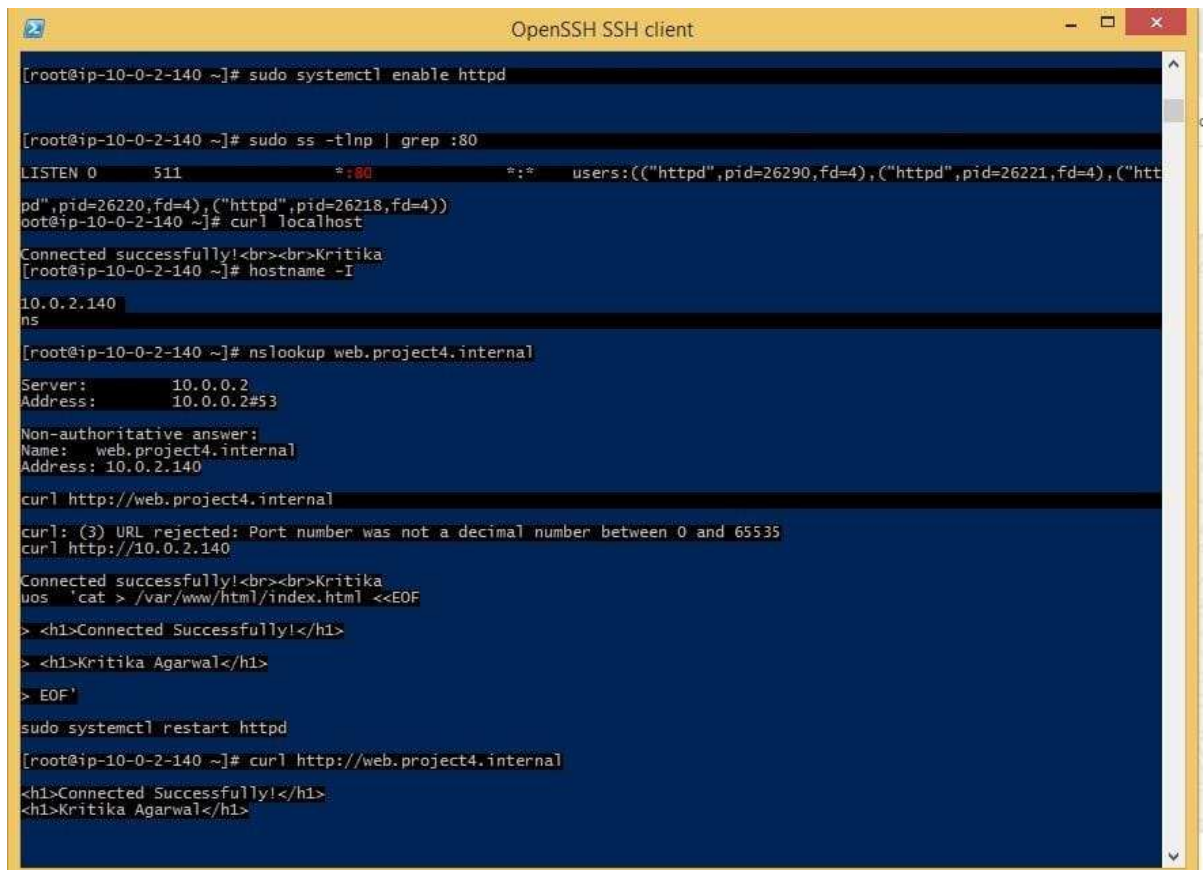
FIG-: INSTALL APACHE

➤ **Step 4-Create Website:**

- Cd/var/www/html
- Sudo nano index.html
- save

➤ **Step 5-Test Website:**

- Browser: http://Public-IP
- Website output



```
[root@ip-10-0-2-140 ~]# sudo systemctl enable httpd

[root@ip-10-0-2-140 ~]# sudo ss -tlnp | grep :80
LISTEN 0      511          *:80         *:*    users:(("httpd",pid=26290,fd=4),("httpd",pid=26221,fd=4),("httpd",pid=26220,fd=4),("httpd",pid=26218,fd=4))
[root@ip-10-0-2-140 ~]# curl localhost
Connected successfully!<br><br>Kritika
[root@ip-10-0-2-140 ~]# hostname -I
10.0.2.140
ns

[root@ip-10-0-2-140 ~]# nslookup web.project4.internal
Server:      10.0.0.2
Address:     10.0.0.2#53

Non-authoritative answer:
Name:   web.project4.internal
Address: 10.0.2.140

curl http://web.project4.internal
curl: (3) URL rejected: Port number was not a decimal number between 0 and 65535
curl http://10.0.2.140
Connected successfully!<br><br>Kritika
uos 'cat > /var/www/html/index.html <<EOF
> <h1>Connected Successfully!</h1>
> <h1>Kritika Agarwal</h1>
> EOF'
sudo systemctl restart httpd

[root@ip-10-0-2-140 ~]# curl http://web.project4.internal
<h1>Connected Successfully!</h1>
<h1>Kritika Agarwal</h1>
```

FIG: CONNECT SUCCESSFULLY

➤ Step 6-Allocate Elastic IP:

- EC2
- Elastic IP
- Allocate
- Associate with Instance.

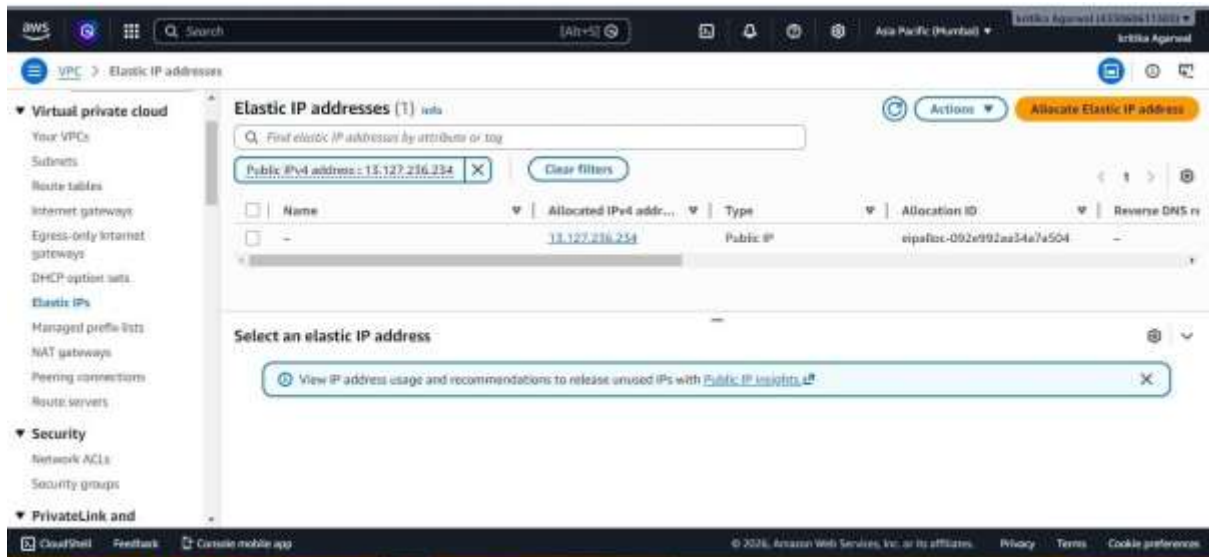


FIG: ELASTIC IP ALLOCATE

➤ Step 7-Route 53 Record:

- Hosted Zone
- Create Record
- Type-A
- Alias-No
- Value-Elastic IP
- Save

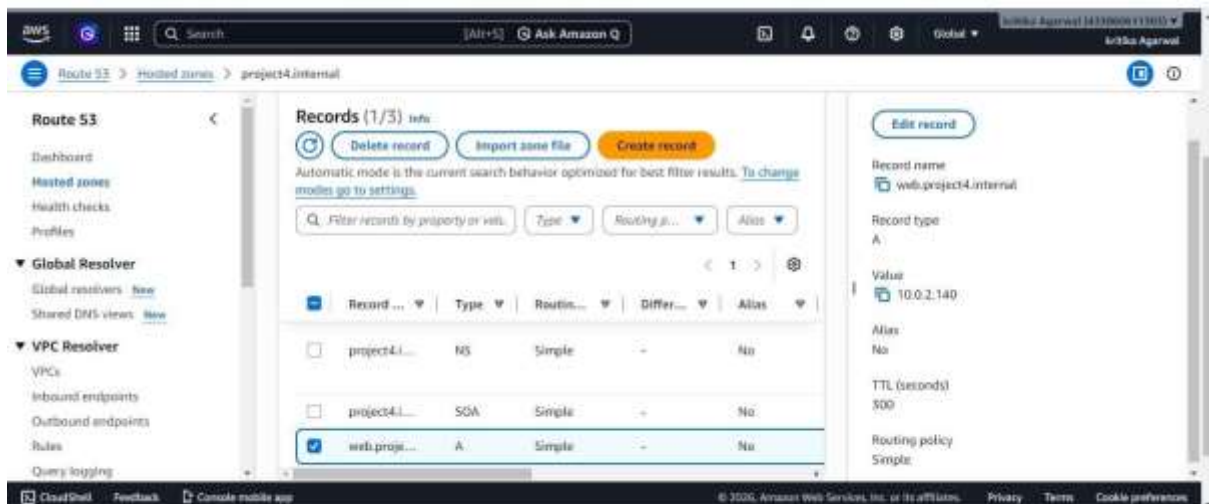


FIG: Route 53 A Record

➤ Step 8-Verification:

- Browser: www.yourdomain.com
- Output: Live website Successfully Opened.



FIG: SUCCESSFULLY VERIFY

5. CONCLUSION:

1. PROJECT SUMMARY-

- Custom Domain successfully opens S3 Static Website.
- Custom Domain successfully opens EC2 Live Website.
- DNS resolution managed using Route 53.
- Elastic IP ensures EC2 IP remains unchanged.

2. LEARNING OUTCOMES-

- Understood Route 53 Hosted Zones and DNS records.
- Learned S3 Static Website Hosting.
- Configured Alias Record for S3.
- Configured A Record for EC2 using Elastic IP.
- Understood why S3 bucket name should match the domain name.

THANK YOU
